

Qatar-2b

R-1017

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_160529 · created 2026-07-30T04:03:05+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457537.68055 +/- 0.00029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.064
Transit depth (Rp/Rs)^2	7.7e-05 +/- 0.011 [%]
Orbital Inclination (inc)	85.21 +/- 1.77
Airmass coefficient 1 (a1)	0.25 +/- 0.15
Airmass coefficient 2 (a2)	0.95 +/- 0.63
Scatter in the residuals of the lightcurve fit is	59.18 %
Best Comparison Star	None
Optimal Aperture	2.65
Optimal Annulus	4.0
Transit Duration (day)	0.0509 +/- 0.0071

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.064 vs 0.16327 (deviation 2.18σ)
Duration fit vs published	0.0509 d vs 0.0767 d (deviation 3.12σ)
Mid-time O-C	-0.5 min
Depth SNR	0.0
Residual scatter	59.18 %

Light curve

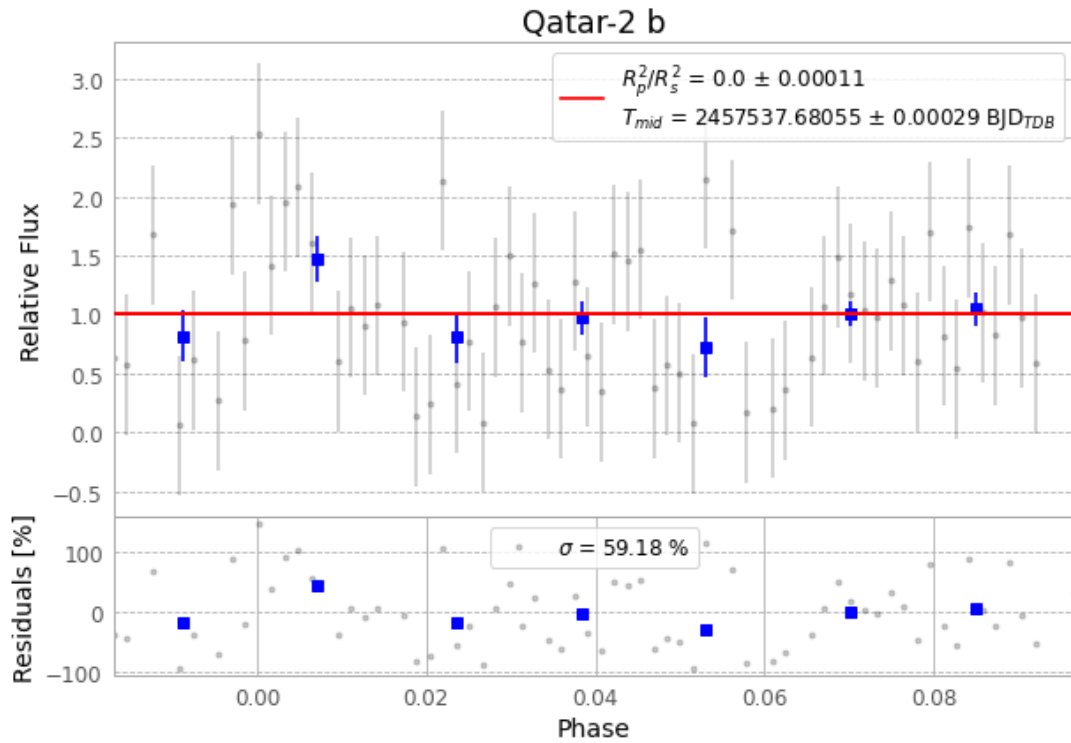


Figure 1 — FinalLightCurve_Qatar-2 b_2016-05-28.png (SHA-256 9045612f5a091fbb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [324, 65], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cbec380bf3a95149...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

74 FITS frames (49 MB), Qatar-2160529033913.FITS → Qatar-2160529071812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-160529.log (7b51d48d6864...), FinalLightCurve_Qatar-2 b_2016-05-28.png (9045612f5a09...), FinalLightCurve_Qatar-2 b_2016-05-28.csv (1a206f3fb3de...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 04:03Z.